

Datasheet for Dataset

Kenya COVID-19 Rapid Response Phone Survey, Wave 4 — as used in “Predicting Household Food Insecurity in Kenya,” Ngao Labs FDSAI Capstone, Group 10. Structured per Gebru et al. (2018), *Datasheets for Datasets*.

Motivation

For what purpose was the dataset created?

To monitor the socioeconomic impact of the COVID-19 pandemic on Kenyan households in near-real time, at a moment when in-person household surveys were impractical. The survey tracks employment, income, food security, coping behaviour, and access to services across successive waves, allowing the World Bank and Kenyan government to observe how households were coping as the pandemic progressed. It was not created for the purpose this capstone puts it to (predictive modelling for humanitarian targeting) — that is a repurposing of a monitoring survey, discussed further under Uses below.

Who created and funded it?

The survey was fielded by the World Bank in partnership with the Kenya National Bureau of Statistics (KNBS), as part of a multi-country COVID-19 Rapid Response Phone Survey series run across several African countries. Data collection and enumeration were contracted out under KNBS oversight.

Composition

What do the instances represent, and how many are there?

Two linked tables. **w4_hh.csv** holds one row per household (4,894 households, 236 columns) covering demographics, housing, assets, agriculture, enterprise activity, transfers, expenditure, the food-security module, and coping strategies. **w4_adult.csv** holds one row per adult household member (10,199 records, 25 columns) covering employment, hours worked, migration status, and education, linked back to the household table by household ID.

Is this a sample, and how was it drawn?

A sample, not a census. Respondents were recruited from two frames, recorded in the *strata* field: a KNBS national phone list, and random digit dialing (RDD). Both frames require phone ownership, which is not universal in Kenya and correlates with income, education, and urban residence — this is the dataset's central representativeness limitation, discussed further below. The full strata label set also includes several refugee-camp and stateless-population codes (Kalobeyei, Kakuma, Dadaab, Shona stateless); none of these are present in the Wave 4 extract used in this project, meaning refugee and stateless populations are entirely absent from this analysis.

Does the data contain a label for this project's task?

No. The survey asks seven food-security questions (worry about food, adults and children going hungry, skipping meals, going a full day without eating) individually; this project constructed the four-class *food_security_status* target ourselves by combining these into a severity score. This is a derived label, not one the survey provides directly, and the bucket boundaries reflect a modelling choice made for this project rather than an official food-security classification standard.

Is any information missing, and does missingness follow a pattern?

Yes. Missingness was screened explicitly in this project (see *notebooks/02_preprocessing.ipynb*); candidate features with more than 70% missingness were dropped rather than imputed. Missingness is not random across all fields — questions about assets, enterprise activity, and transfers are more often unanswered by lower-income and less-educated respondents, meaning the households hardest to characterise are disproportionately the same households most likely to be food insecure.

Collection Process

How was the data collected?

Computer-assisted telephone interviews (CATI), conducted by enumerators contracted under KNBS and World Bank oversight. Respondents answered on behalf of their household. Exact fieldwork dates for Wave 4 are documented on the World Bank Microdata Library's survey page rather than reproduced here, to avoid stating a date range from memory rather than the source documentation.

Was consent obtained, and were respondents compensated?

Verbal informed consent is standard practice for World Bank-fielded phone surveys of this kind; the specific consent script and any compensation arrangement for Wave 4 are documented in the source survey instrument, which this project's authors did not independently verify beyond the anonymised microdata extract itself.

Preprocessing, Cleaning, and Labeling

This project's preprocessing (feature engineering, missingness screening, one-hot encoding, the derived food-security target, and a leakage-safe county-level encoding fit only on the training split) is fully documented and reproducible in *notebooks/01_edata.ipynb* and *notebooks/02_preprocessing.ipynb*. The original, unprocessed survey extract is preserved by the project author but is not redistributed in this repository, in keeping with the source licence (see Distribution below).

Uses

What other tasks could this dataset support?

Employment and income-shock analysis, internal migration studies, and general COVID-19 socioeconomic impact research are all natural fits, since the survey was designed for exactly this kind of monitoring.

What should this dataset not be used for?

It should not be used to identify or make decisions about specific named individuals; the microdata is anonymised and was never intended for individual-level targeting. It should not be treated as representative of households without phone access, or of refugee and stateless populations, both structurally excluded from this extract. It should not be assumed to generalise, unmodified, to a different kind of shock (such as a climate-driven flood event) or to the present day without revalidation — the data reflects Kenyan households' circumstances during a specific pandemic-era window, now several years past.

Distribution

The raw microdata is distributed by the World Bank Microdata Library under a research-purposes-only licence that prohibits redistribution. This project's public repository contains only code, engineered feature definitions, and derived aggregate results (trained model artefacts, evaluation tables, figures) — never the underlying microdata itself. Anyone wishing to reproduce this work must obtain the raw files directly from the source; see *data/raw/data_download.md*.

Maintenance

The source dataset is maintained by the World Bank Microdata Library; this project's authors have no role in maintaining or updating the original survey data. This project's derived code, features, and model artefacts are maintained in this repository by the capstone author and are not scheduled for ongoing updates beyond the capstone submission, though the Responsible AI Statement recommends re-evaluation against more recent data before any operational use.